

Assignment - 13.1

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B - 28

Task 1. Removing Code Duplication Prompt:

Check the Python code. Find repeated code. Move the repeated code into one function. Keep the output the same. Add a docstring for the function.

CODE:

```
def rectangle_details(length, width):  
  
    """Calculate and print area and perimeter of a  
    rectangle."""  
  
    area = length * width  
  
    perimeter = 2 * (length + width)  
  
    print("Area of Rectangle:", area)  
  
    print("Perimeter of Rectangle:", perimeter)  
  
rectangle_details(5, 10)  
  
rectangle_details(7, 12)  
  
rectangle_details(10, 15)
```

OUTPUT:

Area of Rectangle: 50

Perimeter of Rectangle: 30

Area of Rectangle: 84

Perimeter of Rectangle: 38

Area of Rectangle: 150

Perimeter of Rectangle: 50

Task 2. Optimizing Loops Prompt:

Check the Python code with nested loops. Improve the code using set lookup or list comprehension. Keep the logic the same. Show execution time before and after.

CODE:

```
import time

names = ["Alice", "Bob", "Charlie", "David"]
search_names = ["Charlie", "Eve", "Bob"]
start = time.time()

name_set = set(names)

for name in search_names:
    if name in name_set:
        print(name, "is in the list")
    else:
        print(name, "is not in the list")

end = time.time()

print("Execution Time:", end - start)
```

OUTPUT:

Charlie is in the list

Eve is not in the list

Bob is in the list

Execution Time: small decimal value such as 0.00001

Task 3. Extracting Functions Prompt:

Check the Python script. Move repeated calculations into a reusable function. Keep the output the same. Add a docstring.

CODE:

```
def calculate_total(price):  
    """Return total price after tax."""  
    tax_rate = 0.18  
    tax = price * tax_rate  
    total = price + tax  
    return total  
  
print("Total Price:", calculate_total(250))  
print("Total Price:", calculate_total(500))
```

OUTPUT:

Total Price: 295.0
Total Price: 590.0

Task 4. Replace Magic Numbers Prompt:

Check the Python code. Find hardcoded numbers. Create constants at the top of the program. Replace the numbers with these constants.

CODE:

```
PI = 3.14159  
  
RADIUS = 7  
  
area = PI * (RADIUS ** 2)
```

```
circumference = 2 * PI * RADIUS  
  
print("Area of Circle:", area)  
  
print("Circumference of Circle:", circumference)
```

OUTPUT:

Area of Circle: 153.93791

Circumference of Circle: 43.98226

Task 5. Improve Variable Names Prompt:

Check the Python code. Change unclear variable names to meaningful names. Add comments to explain the logic.

CODE:

```
# base of triangle  
  
base = 10  
  
# height of triangle  
  
height = 20  
  
# area calculation  
  
area_of_triangle = base * height / 2  
  
print(area_of_triangle)
```

OUTPUT:

100.0

Task 6. Remove Repeated Conditions Prompt:

Check the Python code with grading conditions. Create a function `calculate_grade(marks)`. Use the function for all grade checks.

CODE:

```
def calculate_grade(marks):  
    """Return grade based on marks."""  
    if marks >= 90:  
        return "Grade A"  
    elif marks >= 75:  
        return "Grade B"  
    else:  
        return "Grade C"  
print(calculate_grade(85))  
print(calculate_grade(72))
```

OUTPUT:

Grade B
Grade C

Task 7. Convert Code into Functions Prompt:

Check the Python script. Split the code into three functions. One for input. One for processing. One for output.

CODE:

```
def get_input():  
    """Read number from user."""  
    number = int(input("Enter number: "))  
    return number  
  
def calculate_square(number):
```

```
        """Return square of number."""
        return number * number
def display_result(result):
    """Print the result."""
    print("Square:", result)
num = get_input()
square = calculate_square(num)
display_result(square)
```

OUTPUT:

Enter number: 5

Square: 25

Task 8. Optimize List Processing Prompt:

Check the Python code that uses loops to process lists. Replace the loop with list comprehension. Keep the same output.

CODE:

```
nums = [1, 2, 3, 4, 5]
squares = [n * n for n in nums]
print(squares)
```

OUTPUT:

[1, 4, 9, 16, 25]